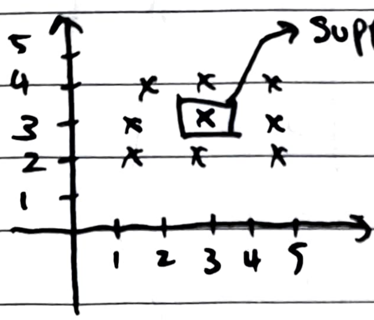
Handwritten representation exercise for <u>DIF</u>.

Input datapoints:

Chosen so that we have this graph. Why? To see if the randomly generated n.n can sucessfully transform the dataset & isolate the point in the middle of the circle



Supposed anomaly (To be isolated).

(But in 3D & in range [-5, 5].).

Datapoints:

1. [0, -5, 5].
2. [-2, -2, 3].
3. [-5, 0, 4].
4. [-3, 3, -2].
5. [0, 5, -3].
6. [2, 3, -1].
7. [5, 0, 0]

[3, -2, -5].

0, 0, 0] → To be isolated

$$D = \begin{bmatrix} 0 & -5 & 5 \\ -2 & -2 & 3 \\ -5 & 0 & 4 \\ -3 & 3 & -2 \\ 0 & 5 & -3 \\ 2 & 3 & -1 \\ 5 & 0 & 0 \\ 3 & -2 & -5 \\ 0 & 0 & 0 \end{bmatrix}$$

Representation 1:

randomly selected values for w, q, p, b are in range [-2, 2].

- Only 1 hidden layer
- $h = \mathrm{ReLU}(W_1 x + b_1)$ → Hidden layer 1
- $o = (W_2 h + b_2)$ → output layer.

$$W_{0,1} = \begin{bmatrix} -1 & 1 & 0 \\ 2 & -1 & 2 \\ -2 & 0 & 1 \end{bmatrix} \quad P_0 = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix} \quad q_0 = \begin{bmatrix} 1 \\ 2 \\ -2 \end{bmatrix}$$

$$W_1 = W_{0,1} \circ P_0 q_0^T = \begin{bmatrix} -1 & 2 & 0 \\ -2 & 2 & 4 \\ -4 & 0 & -4 \end{bmatrix}.$$

$$b_1 = [1, -2, 0].$$

$$P_{output} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix} \quad q_{output} = \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} \quad W_{0,2} = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & -2 \end{bmatrix}$$

$$W_2 = W_{0,2} \circ P_{out} q_{out}^T = \begin{bmatrix} 0 & 1 & -1 \\ 8 & 0 & 2 \\ 2 & -1 & 2 \end{bmatrix}.$$

$$b_2 = [2, 1, -1].$$

Passing the dataset through representation 1:

D = dataset (input)

W_1 = ~~hit~~ hidden layer 1 weights.

b_1 = hidden layer 1 bias.

W_2 = output layer weights.

b_2 = Output layer weights.

O = output dataset

h = data after passing hidden layer

$$h = \mathrm{ReLU}(W_1 D + b_1)$$

$$O = (W_2 h + b_2)$$

$$W_1 D^T + b_1 = \begin{bmatrix} -1 & 2 & 0 \\ -2 & 2 & 4 \\ -4 & 0 & -4 \end{bmatrix} \begin{bmatrix} 0 & -2 & -5 & -3 & 0 & 2 & 5 & 3 & 0 \\ -5 & -2 & 0 & 3 & 5 & 3 & 0 & -2 & 0 \\ 5 & 3 & 4 & -2 & -3 & -1 & 0 & -5 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} -9 & -1 & 6 & 10 & 11 & 5 & -4 & -6 & 1 \\ 8 & 10 & 24 & 2 & -4 & -4 & -12 & -32 & -2 \\ -20 & -4 & 4 & 20 & 12 & -4 & -20 & 8 & 0 \end{bmatrix}$$

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$$h = \mathrm{ReLU}(W_1 D + b_1)$$
$$O = (W_2 h + b_2)$$

$$W_1 D^T + b_1 = \begin{bmatrix} -1 & 2 & 0 \\ -2 & 2 & 4 \\ -4 & 0 & -4 \end{bmatrix} \begin{bmatrix} 0 & -2 & -5 & -3 & 0 & 2 & 5 & 3 & 0 \\ -5 & -2 & 0 & 3 & 5 & 3 & 0 & -2 & 0 \\ 5 & 3 & 4 & -2 & -3 & -1 & 0 & -5 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 & -2 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

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$$h = \mathrm{ReLU}(W_1 D + b_1)$$

$$= \begin{bmatrix} 0 & 0 & 6 & 10 & 11 & 5 & 0 & 0 & 1 \\ 8 & 10 & 24 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 20 & 12 & 0 & 0 & 8 & 0 \end{bmatrix}$$

$$O = \begin{bmatrix} 0 & 1 & -1 \\ 8 & 0 & 2 \\ 2 & -1 & 2 \end{bmatrix} \begin{bmatrix} 0 & 0 & 6 & 10 & 11 & 5 & 0 & 0 & 1 \\ 8 & 10 & 24 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 4 & 20 & 12 & 0 & 0 & 8 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$$

$$O = \begin{bmatrix} 10 & 12 & 22 & -16 & -10 & 2 & 2 & -6 & 2 \\ 1 & 1 & 57 & 121 & 113 & 41 & 1 & 17 & 9 \\ -9 & 11 & -5 & 57 & 45 & 9 & -1 & 15 & 1 \end{bmatrix}$$

Representation 2:

- 2 hidden layer.
- $h_1 = \mathrm{ReLU}(W_1 x + b_1)$ → Hidden layer 1
- $h_2 = \mathrm{ReLU}(W_2 h_1 + b_2)$ → Hidden layer 2
- $o = (W_3 h_2 + b_3)$. → output layer

$$W_{o,1} = \begin{bmatrix} -1 & 1 & 0 \\ 2 & -1 & 2 \\ -2 & 0 & 1 \end{bmatrix} \qquad p_0 = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \qquad q_0 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix}$$

$$W_1 = W_{o,1} \circ (p_0 q_0^T) = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 4 \\ 2 & 0 & -2 \end{bmatrix} \qquad b_1 = [0 \ 1 \ -1].$$

$$W_{o,2} = \begin{bmatrix} 0 & 1 & -1 \\ 2 & 0 & 1 \\ -1 & 1 & -2 \end{bmatrix} \qquad p_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \qquad q_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}.$$

$$W_2 = W_{o,2} \circ (p_1 q_1^T) = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$b_2 = [1 \ -1 \ 2].$$

$$W_{0,3} = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \qquad p_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \qquad q_2 = \begin{bmatrix} 1 \\ -2 \\ 2 \end{bmatrix}$$

$$W_3 = W_{0,3} \circ (p_2 q_2^T) = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$b_3 = [2, 1, -1]$.

Passing the dataset through rep 2:

D = input dataset.

W_i = $i=1$ & 2 ⇒ Hidden layer i weights, $i=3$ = output layer weight

b_i = $i=1$ & 2 ⇒ Hidden layer i bias., $i=3$ = output layer bias.

o = output data

h_i = hidden layer i output.

- $h_1 = ReLU(W_1 D + b_1)$
- $h_2 = ReLU(W_2 h_1 + b_2)$
- $O = W_3 h_2 + b_3$.

$$W_1 D^T + b_1 = \begin{bmatrix} 2 & 2 & 0 \\ 2 & 1 & 4 \\ 2 & 0 & -2 \end{bmatrix} \begin{bmatrix} 0 & -2 & -5 & -3 & 0 & 2 & 5 & 3 & 0 \\ -5 & -2 & 0 & 3 & 5 & 3 & 0 & -2 & 0 \\ 5 & 3 & 4 & -2 & -3 & -1 & 0 & -5 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} -10 & -8 & -10 & 0 & 10 & 10 & 10 & 2 & 0 \\ 16 & 7 & 7 & -10 & -6 & 4 & 11 & -15 & 1 \\ -11 & -11 & -19 & -3 & 5 & 5 & 9 & 15 & -1 \end{bmatrix}.$$

$$h_1 = ReLU(W_1 D^T + b_1) = \begin{bmatrix} 0 & 0 & 0 & 0 & 10 & 10 & 10 & 2 & 0 \\ 16 & 7 & 7 & 0 & 0 & 4 & 11 & 0 & 1 \\ 0 & 0 & 0 & 0 & 5 & 5 & 9 & 15 & 0 \end{bmatrix}.$$

$$W_2h_1 + b_2 = \begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 & 10 & 10 & 10 & 2 & 0 \\ 16 & 7 & 7 & 0 & 0 & 4 & 11 & 0 & 1 \\ 0 & 0 & 0 & 0 & 5 & 5 & 9 & 15 & 0 \end{bmatrix}$$

$$+ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 33 & 15 & 15 & 1 & -4 & 4 & 14 & -14 & 4 \\ -1 & -1 & -1 & -1 & -16 & -16 & -12 & 10 & -2 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$h_2 = \mathrm{ReLU}(W_2h_1 + b_2) = \begin{bmatrix} 33 & 15 & 15 & 1 & 0 & 4 & 14 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$o = W_3h_2 + b_3 = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 33 & 15 & 15 & 1 & 0 & 4 & 14 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 10 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \end{bmatrix}$$

$$+ \begin{bmatrix} 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 & 2 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 & -1 \end{bmatrix}$$

$$o = \begin{bmatrix} 35 & 17 & 17 & 3 & 2 & 6 & 16 & 22 & 6 \\ 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ -34 & -16 & -16 & -2 & -1 & -5 & -15 & -1 & -5 \end{bmatrix}$$